UM-3 model

The universe model UM-3: the author's spacetime of eight dimensions, the fields and where they live, the complete action, and every parameter

Underlying theory: Patrick L. Nash Formulated and verified by Claude (Anthropic)

Model of UM-3, 2026-10-24 Repository once-ere/Dirac_claude, folder universe_model_3

Contents

1 The fold: definition and examples **1**

2 How to read this document **9**
2.1 What this document contains . . . 9
2.2 The status labels . . . 9
2.3 Coordinates, signature and units . . . 9

3 The spacetime **10**

4 The fields and where they live **11**

5 The action **12**
5.1 The sum of all terms . . . 12
5.2 Gravity . . . 13
5.3 dirac16complex . . . 14
5.4 No stabiliser of the bulk . . . 16
5.5 The fold . . . 17
5.6 The Higgs doublet with the ad hoc vacuum . . . 20
5.7 The gauge fields, the quarks and the leptons . . . 21
5.8 The portals of dirac16complex . . . 23

6 The parameters **25**

1 The fold: definition and examples

Definition. The *fold* is the seven-dimensional surface of the author's eight-dimensional spacetime on which the cosine of the angle $z = 6Hx_0$ of the hidden direction vanishes:

$$\mathcal{F} = \big\{X = (x_0, x_1, \ldots, x_7) :\ \cos(6Hx_0) = 0\big\}, \qquad \text{in one period } 0 < z < \pi \text{ of the metric: } z = \frac{\pi}{2}, \quad x_0 = \frac{\pi}{12H} \tag{FD-1}$$

Symbols in (FD-1).

- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status: DERIVED.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.

- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

$X = \{x_0, \ldots, x_7\}$ are the author's coordinates: x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the three extra times; H is the author's constant (an inverse length). The author's metric depends on x_0 only through $\cot^2 z$ and $(\sin^2 z)^{1/6}$, so it repeats with the period $\pi/(6H)$ of x_0, and every period $0 < z < \pi$ contains exactly one surface of $\mathcal{F}$, at $z = \pi/2$: $\mathcal{F}$ is the union of these copies, one per period (example 2), and "the fold" without a qualifier is the copy in the period that contains our world. A point of the fold is labelled by the seven coordinates $x_1, \ldots, x_7$, and the metric restricted to the fold ($\mathrm{d}x_0 = 0$, $z = \pi/2$) is

$$g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N\big|_{\mathcal{F}} = \mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big) \tag{FD-2}$$

Symbols in (FD-2).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

with three space-like and four time-like directions; a_4 and b_4 are the author's two functions of the time x_4. Notation: ln is the natural logarithm, e Euler's number, $\exp(y) = \mathrm{e}^y$, sin, cos and cot = cos / sin the trigonometric functions, tanh the hyperbolic tangent, $|y|$ the absolute value; $f(0^+)$ and $f(0^-)$ are the limits of $f(\sigma)$ as σ approaches 0 from

above and from below.

Why this surface is called a fold. Measured by the proper distance along the hidden direction,

$$\sigma = \operatorname{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0, \qquad \sin z = \mathrm{e}^{-6H|\sigma|} \tag{FD-3}$$

Symbols in (FD-3).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\operatorname{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

(the fold is $\sigma = 0$; the half-period $0 < z < \pi/2$ is $\sigma < 0$, the half-period $\pi/2 < z < \pi$ is $\sigma > 0$; the two ends of the period, the tips $z \to 0$ and $z \to \pi$, lie at $\sigma \to -\infty$ and $\sigma \to +\infty$), the author's metric is

$$g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2 \tag{FD-4}$$

Symbols in (FD-4).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on

the fold and $\mathrm{e}^{-a_4} b_4\, \mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.

- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

It depends on σ only through $|\sigma|$. The function $\mathcal{W}(\sigma) = \mathrm{e}^{-2H|\sigma|}$, which multiplies the squared steps of ordinary space and of the extra times, is the warp factor of the metric (its square root $\mathrm{e}^{-H|\sigma|}$ multiplies the lengths); the time x_4 is not warped. Four properties make the surface $\sigma = 0$ a fold, and together they single it out within the period:

1. *It is a mirror.* The reflection $\sigma \to -\sigma$ (that is $z \to \pi - z$, or $x_0 \to \pi/(6H) - x_0$) maps the metric to itself, and the fold is the only surface of the period that the reflection leaves in place: the two halves of the period are mirror images of each other, joined along the fold, as the two halves of a folded sheet of paper are joined along the crease.
2. *The warp factor is largest on it.* The warp factor $(\sin^2 z)^{1/6} = \mathrm{e}^{-2H|\sigma|}$ equals 1 on the fold and decreases on both sides.
3. *The metric has a kink on it.* The metric is continuous across the fold, but its first derivative along σ jumps. The extrinsic curvature of the surfaces $\sigma = \text{const}$ changes sign across the fold,

$$K^a{}_b = \frac{1}{2}\, g^{ac}\, \partial_\sigma g_{cb}: \qquad K^1{}_1 = K^2{}_2 = K^3{}_3 = K^5{}_5 = K^6{}_6 = K^7{}_7 = -H\,\mathrm{sgn}(\sigma), \qquad K^4{}_4 = 0 \tag{FD-5}$$

Symbols in (FD-5).

- $K^a{}_b$: The extrinsic curvature of a surface $\sigma = \text{const}$ of the hidden direction, $K^a{}_b = \frac{1}{2} g^{ac} \partial_\sigma g_{cb}$ for a, b along the surface (FD-5): $-H\,\mathrm{sgn}(\sigma)$ along ordinary space and the extra times, 0 along the time. Unit: GeV (1/length). Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- c: A frame index (as a). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\, \mathrm{d}^8 X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.

(a, b frame indices along the fold); it jumps by $-2H$ along ordinary space and along the extra times. The Einstein-Lovelock equations of UM-3, $\sum_p \alpha_{(p)} \mathcal{E}_{(p)}{}^a{}_b + \Lambda \delta^a{}_b = \kappa \mathcal{T}^a{}_b$ ($\mathcal{E}_{(p)}$ the Lovelock tensor of order $p = 1, 2, 3$, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ its coupling with the weights $w = (1, 1/20, 1/360)$, $\Lambda = -7H^2$ the cosmological constant, $\kappa = 8\pi G_8$ the gravitational coupling of eight dimensions, $\mathcal{T}^a{}_b$ the energy-momentum tensor, $\delta^a{}_b$ the Kronecker delta), are linear in the second σ-derivatives of the metric, so the jump puts a Dirac delta function $\delta_\mathrm{D}(\sigma)$ into them: the metric solves them only if the fold carries a thin source $\mathcal{S}^a{}_b$, an energy-momentum tensor concentrated on $\sigma = 0$, given by the integral of the equations across the fold (the junction conditions),

$$\kappa\,\mathcal{S}^{a}{}_{b} = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_{p=1}^{3} \alpha_{(p)}\,\mathcal{E}_{(p)}{}^{a}{}_{b} \tag{FD-11}$$

Symbols in (FD-11).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^{a}{}_{b}$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^{a}{}_{b} = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)}\mathcal{E}_{(p)}{}^{a}{}_{b}$ (FD-11, the junction conditions); $\mathcal{S}^{4}{}_{4}$ is minus its energy density, $\mathcal{S}^{1}{}_{1}$ and $\mathcal{S}^{5}{}_{5}$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^{a}{}_{b} = -\frac{32}{5}H\delta^{a}{}_{b}$ at the static block. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.

with $\mathcal{S}^{4}{}_{4} = -$(its energy density), $\mathcal{S}^{1}{}_{1} = \mathcal{S}^{2}{}_{2} = \mathcal{S}^{3}{}_{3}$ its pressure along ordinary space and $\mathcal{S}^{5}{}_{5} = \mathcal{S}^{6}{}_{6} = \mathcal{S}^{7}{}_{7}$ its pressure along the extra times; $\mathcal{S}$ is $\mathcal{T}$ integrated across the fold, an energy per unit volume of the six other directions of the fold (unit GeV^7). On the static block (a_4 and b_4 constant) and on an expanding fold whose extra times are frozen (b_4 proportional to e^{a_4}; $\mathsf{A} = a_4'/H$, $\mathsf{A}_1 = a_4''/H^2$)

$$\begin{aligned}
&\text{static block:}\quad \kappa\mathcal{S}^{a}{}_{b} = -\frac{32}{5}H\,\delta^{a}{}_{b}\quad \text{(a pure tension)},\\
&\text{frozen extra times:}\quad \kappa\mathcal{S}^{1}{}_{1} = -\frac{16}{15}\big(6 + 3\mathsf{A}^2 + 2\mathsf{A}_1\big)H,\quad \kappa\mathcal{S}^{4}{}_{4} = -\frac{16}{5}\big(2 + \mathsf{A}^2\big)H,\\
&\qquad\qquad \kappa\mathcal{S}^{5}{}_{5} = -\frac{4}{5}\big(8 + 8\mathsf{A}^2 + \mathsf{A}^4 + 4\mathsf{A}_1 + \mathsf{A}^2\mathsf{A}_1\big)H
\end{aligned} \tag{FD-12}$$

Symbols in (FD-12).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^{a}{}_{b}$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^{a}{}_{b} = \int_{0^-}^{0^+} \mathrm{d}\sigma \sum_p \alpha_{(p)}\mathcal{E}_{(p)}{}^{a}{}_{b}$ (FD-11, the junction conditions); $\mathcal{S}^{4}{}_{4}$ is minus its energy density, $\mathcal{S}^{1}{}_{1}$ and $\mathcal{S}^{5}{}_{5}$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^{a}{}_{b} = -\frac{32}{5}H\delta^{a}{}_{b}$ at the static block. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); 1 at the author's vacuum, which is not a solution with the corrected weights.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless.

the tension plus matter that obeys the Friedmann equations of general relativity. The fold is therefore not only a place but a physical object: a thin layer that carries energy and momentum (a brane). 4. *The author's coordinate x_0 degenerates on it.* The metric component $g_{x_0x_0} = \cot^2(6Hx_0)$ vanishes on the fold and nowhere else in the period; near the fold the proper distance grows with the square of the coordinate distance,

$$|\sigma| = 3H\Big(x_0 - \frac{\pi}{12H}\Big)^2\Big[1 + 6H^2\Big(x_0 - \frac{\pi}{12H}\Big)^2 + \dots\Big] \tag{FD-6}$$

Symbols in (FD-6).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

so the two coordinate values on either side of $x_0 = \pi/(12H)$ at the same coordinate distance lie at the same proper distance from the fold: the coordinate line of x_0 is folded onto the line of proper distances (a fold map, example 5).

What lives on the fold. In UM-3 the Standard Model (the Higgs doublet with the author's ad hoc vacuum, a time-dependent choice of its vacuum value; the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$; three generations of quarks and leptons and three right-handed neutrinos) and the fold fluid (its tension and the pressure along the extra times that keeps their volume fixed) live on the fold: they are fields at $\sigma = 0$, functions of $x_1, \dots, x_4$ (no field depends on the extra times). Integrated over the extra times, the fold is the four-dimensional universe that we observe, with the metric $\mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in $x_1, \dots, x_4$. dirac16complex (the author's spinor field of eight dimensions) and gravity live in the bulk $\sigma \neq 0$ as well as on the fold; the bound states χ of dirac16complex are the part of it that the fold holds, and the portals (couplings between dirac16complex and Standard-Model fields) act at $\sigma = 0$. In UM-3 files "the fold" means this surface; the examples below use the word for other surfaces with the same properties. The word is not used in its other senses: a set of equal energy levels is called degenerate, and a growth by the factor e is called an e-folding.

Examples of folds that are meaningful for UM-3.

1. *Our fold.* The fold $z = \pi/2$ of the period that contains our world (FD-1): the seven-dimensional world-volume of the Standard Model, and after the integration over the extra times the four-dimensional universe that we observe.
2. *The copies of the fold.* Because the metric repeats with the period $\pi/(6H)$, every surface $x_0 = \pi/(12H) +$ (an integer multiple of $\pi/(6H)$) is a copy of the fold, one per period. Neighbouring copies are separated by a tip (z a multiple of π), where the warp factor $(\sin^2 z)^{1/6}$ vanishes and $g_{x_0x_0} = \cot^2 z$ diverges, and which lies at an infinite proper distance from both: in proper distance every copy has its own complete line $-\infty < \sigma < \infty$. A compact direction of length L would give every field a tower of particles with the masses $2\pi n/L$, $n = 1, 2, \dots$ (a Kaluza-Klein tower); because the copies are infinitely far apart, the period $\pi/(6H)$ of x_0 produces no such tower with the spacing $12H$, and the x_0 mass of the bound states is their binding to the fold.
3. *A folded sheet of paper.* The crease of a folded sheet is a fold in this sense: the two halves are mirror images (property 1) joined along the crease, the sheet is continuous there and its direction changes abruptly (property

3). In UM-3 the direction across the crease is the hidden direction σ, the seven directions along the crease are $x_1, \ldots, x_7$, and the crease is the fold.

4. *A folded function.* The warp factor $\mathrm{e}^{-2H|\sigma|}$ is the simplest folded function: continuous, mirror symmetric, largest at $\sigma = 0$, with a jump of its derivative,

$$\frac{\mathrm{d}}{\mathrm{d}\sigma}\mathrm{e}^{-2H|\sigma|} = -2H\,\mathrm{sgn}(\sigma)\,\mathrm{e}^{-2H|\sigma|}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}\sigma^2}\mathrm{e}^{-2H|\sigma|} = 4H^2\,\mathrm{e}^{-2H|\sigma|} - 4H\,\delta_{\mathrm{D}}(\sigma) \tag{FD-7}$$

Symbols in (FD-7).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- δ_{D}: The Dirac delta function: $\int \mathrm{d}\sigma\,\delta_{\mathrm{D}}(\sigma)F(\sigma) = F(0)$. Unit: 1/length (for $\delta_{\mathrm{D}}(\sigma)$). Status: STANDARD.

whose delta function is the origin of the thin source of property 3. Every metric function of FD-4 depends on σ through $|\sigma|$ and is folded at $\sigma = 0$ in this sense. 5. *A fold map.* In the mathematics of smooth maps the fold (Whitney's fold singularity) is the map $u \mapsto u^2$ of a line onto a half-line: it folds the line at $u = 0$, two points $\pm u$ land on the same point, and the derivative of the map vanishes on the fold. Near the fold the map from the author's coordinate x_0 to the distance $|\sigma|$ from the fold is of this kind (FD-6), and $g_{x_0x_0} = \cot^2 z$, the square of the derivative $\mathrm{d}\sigma/\mathrm{d}x_0$, vanishes there. 6. *The brane of Randall and Sundrum.* The five-dimensional metric

$$\mathrm{d}y_{\mathrm{RS}}^2 + \mathrm{e}^{-2k_{\mathrm{RS}}|y_{\mathrm{RS}}|}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 - \mathrm{d}x_4^2\big), \qquad \lambda_{\mathrm{RS}} = \frac{6k_{\mathrm{RS}}}{\kappa_5}, \qquad \Lambda_{\mathrm{RS}} = -\frac{6k_{\mathrm{RS}}^2}{\kappa_5} \tag{FD-8}$$

Symbols in (FD-8).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- y_{RS}: The proper coordinate across the brane of the five-dimensional model of Randall and Sundrum (FD-8, an example, not UM-3). Unit: length. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- k_{RS}: The inverse curvature length of the bulk of the model of Randall and Sundrum: the warp factor of the brane is $\mathrm{e}^{-2k_{\mathrm{RS}}|y_{\mathrm{RS}}|}$ (FD-8, an example, not UM-3). Unit: GeV (1/length). Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- λ_{RS}: The tension (energy per unit volume of ordinary space) of the brane of the model of Randall and Sundrum, $\lambda_{\mathrm{RS}} = 6k_{\mathrm{RS}}/\kappa_5$ (FD-8, an example, not UM-3). Unit: GeV^4. Status: STANDARD.
- κ_5: The gravitational coupling of five dimensions in the model of Randall and Sundrum, $\kappa_5 = 8\pi G_5$ (FD-8, an example, not UM-3). Unit: GeV^{-3}. Status: STANDARD.
- Λ_{RS}: The vacuum energy density of the bulk of the model of Randall and Sundrum, $\Lambda_{\mathrm{RS}} = -6k_{\mathrm{RS}}^2/\kappa_5$ (FD-8, an example, not UM-3). Unit: GeV^5. Status: STANDARD.

has a fold at $y_{\mathrm{RS}} = 0$ with the properties 1 to 3 (its coordinate y_{RS} is already the proper distance, so property 4 does not arise); Einstein's equations require there a brane of the tension λ_{RS} (Israel's junction condition) and in the bulk the vacuum energy Λ_{RS}, and the bulk is otherwise empty. The fold of UM-3 is of the same kind: its static block is a vacuum of the bulk with a fold of pure tension (FD-12). It differs in that the time x_4 is not warped, three extra times are warped together with ordinary space, the Lovelock terms add to the junction conditions, and the

space is not cut at the brane: Randall and Sundrum identify the two sides of their brane (one physical side), while the fold of UM-3 has two distinct sides (example 7). 7. *A particle bound to a thin layer.* A particle held by an attractive thin layer has a wave function that falls off exponentially with the distance on both sides, and the layer makes its derivative jump (for the first-order Dirac equation, the wave function itself). In UM-3 the particles χ of dark matter are the states of dirac16complex bound to the fold; at rest and without their self-interaction (the field is $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{-\mathrm{i}\omega_\mathrm{b} x_4}$)

$$\begin{aligned} &\chi(\sigma) = \mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^+) \quad (\sigma > 0), \qquad \chi(\sigma) = \mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^-) \quad (\sigma < 0), \\ &\chi(0^+) = \exp\Big(\gamma^{(0)}\big(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R}\big)\Big)\chi(0^-), \qquad \omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2} \end{aligned} \tag{FD-9}$$

Symbols in (FD-9).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_R: The author's right-handed chiral projector, $P_\mathrm{R} = (\mathbf{1} + \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8 = T_0 T_1 \cdots T_7$ (the author's T16^A[8]) $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; $\Gamma_8^2 = \mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

where μ_L, μ_R are the strengths of the fold for the left- and right-handed halves of dirac16complex (set by the Standard-Model fields on the fold) and $q_\mathrm{b} = -m_\Psi(\mu_\mathrm{L} + \mu_\mathrm{R})\tanh\varsigma/(2\varsigma)$, $\varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}$ ($q_\mathrm{b} = -m_\Psi\mu_\mathrm{L}/2$ for $\mu_\mathrm{R} = 0$). The profile is not a function of $|\sigma|$: it jumps at the fold, $\chi(0^+) \neq \chi(0^-)$. 8. *The fold as a mirror for dirac16complex.* The reflection across the fold acts on the profiles $\chi(\sigma)$ in two ways,

$$(\mathsf{R}_1\chi)(\sigma) = T_0 T_1 T_2 T_3\,\chi(-\sigma), \qquad (\mathsf{R}_2\chi)(\sigma) = T_0 T_5 T_6 T_7\,\chi(-\sigma) \tag{FD-10}$$

Symbols in (FD-10).

- R_1: The reflection across the fold acting on a profile of dirac16complex together with the reflection of ordinary space, $(\mathsf{R}_1\chi)(\sigma) = T_0 T_1 T_2 T_3\,\chi(-\sigma)$ (FD-10): a symmetry of dirac16complex, of gravity and of the mean field of the gas of the particles χ; the bound states of Krein charge -1 are even, those of Krein charge $+1$ odd. Unit: dimensionless. Status: DERIVED.
- R_2: The reflection across the fold acting on a profile of dirac16complex together with the reflection of the extra times, $(\mathsf{R}_2\chi)(\sigma) = T_0 T_5 T_6 T_7\,\chi(-\sigma)$ (FD-10): it maps the bound states of one Krein charge onto those of the other. Unit: dimensionless. Status: DERIVED.
- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit:

GeV^2 (with the V_* normalisation). Status: DERIVED.

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- T_a, T^a: The author's eight real 16×16 Dirac matrices T16ˆA[a] of 4+4.nb ($a = 0, \ldots, 7$; $T_0, \ldots, T_7$), with $\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.

(T_a the author's Dirac matrices; for states at rest R_1 is the reflection across the fold combined with the reflection of ordinary space, R_2 the reflection across the fold combined with the reflection of the extra times). The Krein charge of a state is its share of the conserved charge of dirac16complex, which is indefinite: a state can carry the charge -1 or $+1$. R_1 is a symmetry of dirac16complex, of gravity and of the mean field of a gas of the particles χ (the average field that the particles of the gas create and feel): the four bound states of Krein charge -1 (the particles χ) are even under it and the four of Krein charge $+1$ are odd, so the mean field of the gas cannot mix the two signs of the indefinite charge. R_2 maps the bound states of one Krein charge onto those of the other and makes the two sets degenerate without the self-interaction. 9. *The planes of Horava and Witten.* In M-theory (the eleven-dimensional theory whose limits include the superstring theories) on an interval (a circle whose points are identified with their mirror images), the two ten-dimensional planes left in place by the reflection are mirror surfaces (property 1); each carries one E_8 gauge group while gravity fills the bulk. As in UM-3, the gauge fields live on the mirror surface and gravity in the bulk; unlike UM-3, the two sides of each plane are identified (one physical side).

What is not a fold.

- *The tips* $z = 0$ and $z = \pi$ (mod π): the mirror exchanges them within a period, but there the warp factor vanishes, $g_{x_0x_0}$ diverges, and they lie at an infinite proper distance; they carry no thin source.
- *A smooth maximum of a warp factor*, such as $1/\cosh^2(H\sigma)$: mirror symmetric and largest at $\sigma = 0$, but without a kink and without a thin source.

2 How to read this document

2.1 What this document contains

This document defines the universe model UM-3, the revision of 2026-10-24 ordered by the author, with the Lovelock weights and the cosmological constant corrected at his order of 2026-10-25 so that gravitons travel at the speed of light: the spacetime and its metric, the fields and where they live, the complete action from which every field equation follows, and every parameter with its value and its origin. It is self-contained: every equation is written out (the terms of the action carry the tags IN-1 to IN-16 and the metric the tag FE-27, the same tags as in every UM-3 document), and every symbol of every display equation is defined right after it.

2.2 The status labels

Every equation and every parameter carries one of the labels **AUTHOR** (from the author: his notebooks `4+4.nb` and `Dirac_Einstein-Lovelock_4+4_Nash.nb` and his orders of 2026-10-24 and 2026-10-25; the corrected values of the Lovelock weights and of the cosmological constant that the second order requires are DERIVED), **RECORD** (from the record of this repository: the Lagrangian of dirac16complex with its canonical spin connection and its contact self-interaction, the Kohn-Sham treatment), **STANDARD** (textbook physics or a published measurement), **ASSUMED** (a choice of UM-3 without derivation), **INVENTED** (a new interaction of UM-3: the portals of dirac16complex), **DERIVED** (computed and verified in UM-3), **ESTIMATE** (an order of magnitude) and **OPEN** (not settled).

2.3 Coordinates, signature and units

The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$, always with the hidden direction first: x_0 is the hidden space-like coordinate, x_1, x_2, x_3 ordinary space, x_4 the time, and x_5, x_6, x_7 three extra time-like coordinates. The signature is $(+,+,+,+,-,-,-,-)$ in this order. Units are natural, $\hbar = c = 1$. A prime on a function of the time x_4 alone is its derivative with respect to x_4.

3 The spacetime

$$
\begin{aligned}
g_{MN} &= \mathrm{diag}\Big(\cot^2 z,\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ -1,\\
&\qquad -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6}\Big), \qquad z = 6Hx_0,\\
\sigma &= \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0,\\
g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N &= \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2
\end{aligned}
\tag{FE-27}
$$

Symbols in (FE-27).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The author's metric, with his two functions of the time $a_4(x_4)$ and $b_4(x_4)$ (b_4 is never identically 1, by the author's

order) and his constant H. The metric depends on x_0 only through $z = 6Hx_0$; one period $0 < z < \pi$ of x_0 is, in the proper coordinate σ of the hidden direction, the whole line $-\infty < \sigma < \infty$ (DERIVED). In that coordinate ordinary space and the extra times are warped by $\mathrm{e}^{-H|\sigma|}$, which has a kink at $\sigma = 0$ ($z = \pi/2$): this seven-dimensional surface is the fold. The rest of spacetime is the bulk. Status: AUTHOR (the metric), DERIVED (the chart).

4 The fields and where they live

field	symbol	what it is	lives in	couples to
Einstein-Lovelock gravity	g_{MN}	the metric of eight dimensions, with the Lovelock weights $1, 1/20, 1/360$ and $\Lambda = -7H^2$ (corrected at the author's order of 2026-10-25)	bulk and fold	every field
dirac16complex	Ψ	the author's 16-component complex spinor field, with left- and right-handed halves Ψ_{L}, Ψ_{R}	the bulk	gravity (canonical spin connection), itself, every Standard-Model field at the fold; only Ψ_{L} to the Higgs doublet
Higgs doublet	Φ	the scalar doublet with $V = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$, $\lambda > 0$, $\mu^2 < 0$, expanded about the ad hoc vacuum $(\mathcal{A}, v + h)/\sqrt{2}$	the fold	gravity, W, B, the fermions, Ψ_{L}
lepton doublets	ℓ_i	$(\nu_{\mathrm{L}i}, e_{\mathrm{L}i})$, three generations	the fold	gravity, W, B, Φ, Ψ_{L}, Ψ_{R}
right-handed charged leptons	$e_{\mathrm{R}i}$	electron, muon, tau	the fold	gravity, B, Φ, Ψ_{L}, Ψ_{R}
right-handed neutrinos	$\nu_{\mathrm{R}i}$	three gauge singlets: Dirac neutrino masses (accepted by the author, 2026-10-24 17:32)	the fold	gravity, Φ, Ψ_{L}, Ψ_{R}
electroweak gauge fields	W^1, W^2, W^3, B	the gauge fields of $SU(2)_{\mathrm{L}}$ and $U(1)_Y$	the fold	gravity, Φ, the fermions, Ψ_{L}, Ψ_{R}
quarks	Q_i, $u_{\mathrm{R}i}$, $d_{\mathrm{R}i}$	three generations of doublets and singlets	the fold	gravity, gluons, W, B, Φ, Ψ_{L}, Ψ_{R}
gluons	G^A	the gauge field of $SU(3)_{\mathrm{C}}$	the fold	gravity, quarks, Ψ_{L}, Ψ_{R}
fold fluid	$\mathcal{T}_{\mathrm{ff}}$	the part of the thin source of the fold not supplied by the other fields: a tension, which the static block requires, and a pressure along the extra times that holds their volume fixed	the fold	gravity
induced gravity	κ_{ind}	an Einstein-Hilbert term that the fields of the fold may induce; optional, because with the corrected weights the fold has an Einstein-Hilbert term of its own	the fold	gravity

The Standard Model lives on the fold, because in the bulk every species would move with its own limiting speed (its square is the inverse warp factor averaged over its profile) and three neutrino flavours with three profiles would oscillate with phases that measurements exclude; on the fold every field has the one light cone of the fold (the limiting speeds are DERIVED; the place is ASSUMED on that basis). dirac16complex, the author's field of eight dimensions, lives in the bulk and couples to the Standard Model at the fold. No field depends on the extra times x_5, x_6, x_7: a field with a momentum along them is tachyonic (its frequency squared falls with that momentum, because the extra times are time-like), and the bound particles of dirac16complex with such a momentum leak into the bulk; the restriction is consistent but unstable, and UM-3 needs a gauge principle that removes the momentum

along the extra times of every field. With the author's original Lovelock weights the static bulk needed a medium of its own, the stabiliser (found and named by Claude, Anthropic, on 2026-10-24); with the corrected weights it is zero, and the bulk contains no field but dirac16complex. Status: AUTHOR (the list of fields), ASSUMED (the additions, the places and the restriction), DERIVED (the tachyons, the absence of the stabiliser), OPEN (the gauge principle).

5 The action

All field equations of UM-3 follow from one action by varying it with respect to every field.

5.1 The sum of all terms

$$\begin{aligned} S_{\text{tot}} &= S_{\text{EL}} + S_{\Psi} + S_{\text{fold}}, \\ S_{\text{fold}} &= \int \mathrm{d}^8X \, \sqrt{|g|} \, \delta_{\text{D}}(\sigma) \, \frac{1}{V_*} \left(\mathcal{L}_{\text{IG}} + \mathcal{L}_{\text{ff}} + \mathcal{L}_{\text{H}} + \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{f}} + \mathcal{L}_{\text{Y}} + \mathcal{L}_{\text{P}} \right) \end{aligned} \tag{IN-1}$$

Symbols in (IN-1).

- S_{tot}: The action of UM-3 (the integral of the Lagrangian density over spacetime); its variation with respect to every field gives the field equations. Unit: dimensionless ($\hbar = 1$). Status: ASSUMED.
- S_{EL}: The Einstein-Lovelock action of gravity in eight dimensions. Unit: dimensionless. Status: AUTHOR.
- S_{Ψ}: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- S_{fold}: The action of the fold: the Standard Model, the portals of dirac16complex, the fold fluid and the Einstein-Hilbert term that the fields of the fold induce, all at $\sigma = 0$. Unit: dimensionless. Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0 \, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN} \, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|} \, \mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- δ_{D}: The Dirac delta function: $\int \mathrm{d}\sigma \, \delta_{\text{D}}(\sigma) F(\sigma) = F(0)$. Unit: 1/length (for $\delta_{\text{D}}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z| \, \mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\text{t}}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\text{t}} = \int \mathrm{d}x_5 \, \mathrm{d}x_6 \, \mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{L}_{\text{IG}}$: The Lagrangian of the Einstein-Hilbert term of the four-dimensional metric of the fold that the quantum fluctuations of the fields of the fold induce; it adds to the Einstein-Hilbert term that the Lovelock terms give the fold and does not change the speed of gravitational waves. Unit: GeV^4. Status: ESTIMATE.
- $\mathcal{L}_{\text{ff}}$: The Lagrangian of the fold fluid (only its energy-momentum tensor $\mathcal{T}_{\text{ff}}$ enters the equations). Unit: GeV^4. Status: ASSUMED.
- $\mathcal{L}_{\text{H}}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_{\text{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\text{f}}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\text{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_{\text{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The action has three parts: Einstein-Lovelock gravity in eight dimensions, dirac16complex in the bulk, and the fold, whose terms are confined to $\sigma = 0$ by the Dirac delta of the proper hidden coordinate; with the corrected Lovelock weights the bulk needs no other source (IN-7). Status: ASSUMED.

5.2 Gravity

$$S_{\mathrm{EL}} = \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)} - 2\Lambda\Big), \qquad \kappa = 8\pi G_8,$$
$$\alpha_{(p)} = w_{(p)}\,H^{2-2p}, \qquad w_{(1)} = 1, \quad w_{(2)} = \frac{1}{20}, \quad w_{(3)} = \frac{1}{360}, \qquad \Lambda = -7H^2 \tag{IN-2}$$

Symbols in (IN-2).

- S_{EL}: The Einstein-Lovelock action of gravity in eight dimensions. Unit: dimensionless. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant, $\Lambda = -7H^2$: with the corrected weights the value for which the static block is a vacuum (its Lovelock tensors add to $7H^2$ in every direction). The author's original value was $\Lambda = -10H^2$ (he writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p}w_{(p)}\mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-7H^2$; the author's original $-10H^2$. Status: AUTHOR (the term), DERIVED (the value).
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa = 8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.
- $w_{(p)}$: The Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/20$, $w_{(3)} = 1/360$: the only weights for which gravitational waves travel at the speed of light in the bulk and on the fold (both coefficients of the bulk wave, $1 - 40w_{(2)} + 360w_{(3)}$ and $1 - 24w_{(2)} + 72w_{(3)}$, vanish). The author's original weights $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) made bulk gravitons faster than light; they were corrected at the author's order of 2026-10-25. Unit: dimensionless. Value: $(1, 1/20, 1/360)$; the author's original $(1, 1/32, 1/1152)$. Status: AUTHOR (the form), DERIVED (the values).
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

$$\mathcal{L}_{(p)} = \frac{1}{2^p}\,\delta^{M_1N_1\cdots M_pN_p}_{P_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p},$$
$$\mathcal{L}_{(1)} = R, \qquad \mathcal{L}_{(2)} = R^2 - 4R_{MN}R^{MN} + R_{MNPQ}R^{MNPQ} \tag{IN-3}$$

Symbols in (IN-3).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p (a scalar built from p Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n\times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.

- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor (convention $[\nabla_P, \nabla_Q]V^M = R^M{}_{NPQ}V^N$), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\text{length}^2$. Status: STANDARD.

Einstein-Lovelock gravity in the author's form (his notebook, In[134]) with corrected values: the Einstein-Hilbert, the Gauss-Bonnet and the cubic Lovelock densities, built from the Riemann tensor with the generalised Kronecker delta. The author's notebook had the weights $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ and the cosmological constant $\Lambda = -10H^2$; with them a gravitational wave in the bulk moved with the local speed $\sqrt{5}$ times that of light and the static bulk was not a vacuum. A gravitational wave in the bulk has, for any weights, the time-kinetic coefficient $1 - 40w_{(2)} + 360w_{(3)}$ and the gradient coefficient $1 - 24w_{(2)} + 72w_{(3)}$ (in units of their Einstein values); the wave of the fold travels at the speed of light only if the first vanishes, and with the first zero and the second not the bulk graviton would have an infinite local speed. Both vanish only for $w_{(2)} = 1/20$, $w_{(3)} = 1/360$, the weights of UM-3 at the author's order of 2026-10-25 ("Gravitons must travel at c"). With them the static block is a vacuum for $\Lambda = -7H^2$ (DERIVED). Their variation contains no derivative of the metric higher than the second, so the field equations are of second order and a thin source at the fold produces a finite jump of first derivatives. Status: AUTHOR, STANDARD.

5.3 dirac16complex

$$S_\Psi = \int \mathrm{d}^8X\, \sqrt{|g|}\, \Big[\frac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m_\Psi\, \bar\Psi\Psi - \frac{\lambda_\Psi}{2}\big(\bar\Psi\Psi\big)^2\Big] \tag{IN-4}$$

Symbols in (IN-4).

- S_Ψ: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\text{GeV}^{7/2}$. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.

$$\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}, \qquad \eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1), \qquad T^a = \eta^{ab}T_b, \qquad \gamma^M = \mathbf{e}_a{}^M\, T^a,$$
$$\bar\Psi = \Psi^\dagger\sigma_{16}, \qquad \sigma_{16} = T_0T_1T_2T_3, \qquad \Gamma_8 = T_0T_1T_2T_3T_4T_5T_6T_7, \tag{IN-5}$$
$$P_\mathrm{L} = \frac{\mathbf{1}-\Gamma_8}{2}, \qquad P_\mathrm{R} = \frac{\mathbf{1}+\Gamma_8}{2}, \qquad \Psi_\mathrm{L} = P_\mathrm{L}\Psi, \qquad \Psi_\mathrm{R} = P_\mathrm{R}\Psi$$

Symbols in (IN-5).

- $T_a,\ T^a$: The author's eight real 16×16 Dirac matrices T16^A[a] of 4+4.nb ($a = 0,\dots,7$; $T_0,\dots,T_7$), with $\{T_a,T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0,1,\dots,7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a,b\in\{1,\dots,7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$ (eta4488 of 4+4.nb); η^{ab} is its inverse (the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1},\ \mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0,x_1,\dots,x_7$ for a tensor of eight dimensions, $x_1,\dots,x_7$ for a tensor of the fold, $x_1,\dots,x_4$ for a four-dimensional tensor). Status: STANDARD.
- $\gamma^M,\ \gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0,x_1,\dots,x_7$; a repeated index is summed. Status: STANDARD.
- $\mathbf{e}_a{}^M,\ \mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- $\Psi,\ \bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8 = T_0T_1\cdots T_7$ (the author's T16^A[8]) $= \mathrm{diag}(-\mathbf{1}_8,\mathbf{1}_8)$; $\Gamma_8^2 = \mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1}-\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_R: The author's right-handed chiral projector, $P_\mathrm{R} = (\mathbf{1}+\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Ψ_L: The left-handed half of dirac16complex, $\Psi_\mathrm{L} = P_\mathrm{L}\Psi$ (the upper eight components in the basis of 4+4.nb). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- Ψ_R: The right-handed half of dirac16complex, $\Psi_\mathrm{R} = P_\mathrm{R}\Psi$ (the lower eight components). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

$$D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \qquad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\,\Omega_M, \qquad \Omega_M = \frac{1}{2}\,\varpi_{Mab}\,\Sigma^{ab}, \qquad \Sigma^{ab} = \frac{1}{4}\big[T^a, T^b\big],$$
$$\varpi_{Mab} = \mathbf{e}_a{}^N\big(\partial_M\mathbf{e}_{bN} - \Gamma^P{}_{MN}\,\mathbf{e}_{bP}\big) \tag{IN-6}$$

Symbols in (IN-6).

- $D_M,\ D_\mu$: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0,x_1,\dots,x_7$; a repeated index is summed. Status: STANDARD.

- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- Ω_M: The spin connection acting on dirac16complex, $\Omega_M = \frac{1}{2}\varpi_{Mab}\Sigma^{ab}$. Unit: 1/length. Status: RECORD.
- ϖ_{Mab}: The canonical (Levi-Civita) spin connection of the vielbein, antisymmetric in a, b. Unit: 1/length. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu\mathbf{e}_b{}^\nu\Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- T_a, T^a: The author's eight real 16×16 Dirac matrices T16ˆA[a] of 4+4.nb ($a = 0, \ldots, 7$; $T_0, \ldots, T_7$), with $\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Gamma^P{}_{MN}$: The Christoffel symbols of the metric g_{MN} (the Levi-Civita connection). Unit: 1/length. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

dirac16complex with the kinetic term in the symmetric form (real for the author's real matrices), the mass m_Ψ and the contact self-interaction λ_Ψ (RECORD); the author's eight real 16×16 Dirac matrices, the Dirac adjoint and the chiral projectors (AUTHOR); the canonical spin connection, which couples dirac16complex to gravity and is never identically zero for the author's metric ($\gamma^M\Omega_M = -3H\,\mathrm{sgn}(\sigma)\gamma^{(0)} + \frac{3}{2}(b_4'/b_4)\gamma^{(4)}$, DERIVED; here $\gamma^{(0)} = T^0$ and $\gamma^{(4)} = T^4$ are the author's matrices along the hidden direction and along the time, σ the proper coordinate of the hidden direction with the fold at $\sigma = 0$, and sgn the sign function).

5.4 No stabiliser of the bulk

$$\kappa\,\mathcal{T}_{\mathrm{stab}}{}^a{}_b = \Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{E}_{(p)}{}^a{}_b + \Lambda\,\delta^a{}_b\Big)\Big|_{\text{static block}} = 0: \tag{IN-7}$$
$$(15 - 9 + 1 - 7)H^2 = 0 \text{ (warped directions)}, \qquad (21 - 21 + 7 - 7)H^2 = 0 \text{ (the time)}$$

Symbols in (IN-7).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\mathrm{stab}}{}^a{}_b$: The energy-momentum tensor of the stabiliser of the bulk (frame components): the source that a static bulk of the author's metric required with the author's original Lovelock weights, found and named by Claude (Anthropic) on 2026-10-24; with the corrected weights and $\Lambda = -7H^2$ it is zero (IN-7). Unit: GeV^8. Value: 0 with the corrected weights. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number; in the Kohn-Sham equations a second label of a Kohn-Sham orbital (as o, with δ_{op} the Kronecker delta of the two labels). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^M{}_N$: The Lovelock tensor of order p: the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to the metric ($\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor); with frame indices its frame components. Unit: 1/length^{2p}. Status: STANDARD.

- Λ: The eight-dimensional cosmological constant, $\Lambda = -7H^2$: with the corrected weights the value for which the static block is a vacuum (its Lovelock tensors add to $7H^2$ in every direction). The author's original value was $\Lambda = -10H^2$ (he writes the pure number $\Lambda_{\mathrm{N}} = 10$ in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$). Unit: GeV^2. Value: $-7H^2$; the author's original $-10H^2$. Status: AUTHOR (the term), DERIVED (the value).
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1 \cdots M_n}_{N_1 \cdots N_n}$, the determinant of the $n \times n$ matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)} H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.

The stabiliser of the bulk was the medium that a static bulk of the author's metric required with the author's original weights and cosmological constant: an energy-momentum tensor with the negative energy density $\kappa\rho_{\mathrm{s}} = -H^2/16$ and the pressure $\kappa p_{\mathrm{s}} = -5H^2/16$ in every direction but the time, which violated the weak and the null energy condition and which only a field with negative kinetic energy (a ghost) reproduced. Claude (Anthropic) found and named it on 2026-10-24 while deriving the field equations of UM-3. It was a consequence of the original weights: with the corrected weights the left-hand side of the bulk equations at the static block vanishes in every direction (the display, order by order: Einstein, Gauss-Bonnet, cubic and the cosmological constant), the static bulk is a vacuum, and so is the bulk of an expanding fold with frozen extra times. UM-3 has no stabiliser. Status: DERIVED.

5.5 The fold

$$\int \mathrm{d}^8X \, \sqrt{|g|} \, \delta_{\mathrm{D}}(\sigma) \, \frac{1}{V_*} \, \mathbb{F} = \int \mathrm{d}^4x \, \sqrt{-\hat{h}} \, \mathcal{Z} \, \mathbb{F}\big|_{\sigma=0}, \qquad \mathrm{d}^4x = \mathrm{d}x_1 \, \mathrm{d}x_2 \, \mathrm{d}x_3 \, \mathrm{d}x_4,$$
$$\hat{h}_{\mu\nu} = \mathrm{diag}\big(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1\big), \qquad \sqrt{-\hat{h}} = \mathrm{e}^{3a_4}, \qquad \mathcal{Z} = \frac{V_{\mathrm{t}} \big(\mathrm{e}^{-a_4} b_4\big)^3}{V_*}, \qquad V_{\mathrm{t}} = \int \mathrm{d}x_5 \, \mathrm{d}x_6 \, \mathrm{d}x_7 \tag{IN-8}$$

Symbols in (IN-8).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0 \, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN} \, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|} \, \mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- δ_{D}: The Dirac delta function: $\int \mathrm{d}\sigma \, \delta_{\mathrm{D}}(\sigma) F(\sigma) = F(0)$. Unit: 1/length (for $\delta_{\mathrm{D}}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z| \, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4} b_4)^3$ (with $V_{\mathrm{t}} = \int \mathrm{d}x_5 \, \mathrm{d}x_6 \, \mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4} b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathbb{F}$: Any function of the fields at the fold (used to state a rule of integration). Status: STANDARD.
- x: The coordinates of the fold: x_i ($i = 1, 2, 3$) of ordinary space, x_4 the time, x_r ($r = 5, 6, 7$) the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1 \, \mathrm{d}x_2 \, \mathrm{d}x_3 \, \mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- V_t: The coordinate volume of the (compact) extra times, $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

$$\mathcal{L}_\mathrm{IG} = \frac{\hat{R}}{2\kappa_\mathrm{ind}} \tag{IN-9}$$

Symbols in (IN-9).

- $\mathcal{L}_\mathrm{IG}$: The Lagrangian of the Einstein-Hilbert term of the four-dimensional metric of the fold that the quantum fluctuations of the fields of the fold induce; it adds to the Einstein-Hilbert term that the Lovelock terms give the fold and does not change the speed of gravitational waves. Unit: GeV^4. Status: ESTIMATE.
- $\hat{R}$: The scalar curvature of the metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- κ_ind: The coupling of the Einstein-Hilbert term $\hat{R}/(2\kappa_\mathrm{ind})$ that the quantum fluctuations of the fields of the fold may induce, $\kappa_\mathrm{ind} = 8\pi G_\mathrm{ind}$; it adds to the gravity of the fold, $1/(8\pi G_\mathrm{N}) = V_*/\kappa_7 + 1/\kappa_\mathrm{ind}$. Unit: GeV^{-2}. Value: not fixed by UM-3. Status: ESTIMATE.

$$\mathcal{T}_\mathrm{ff}{}^a{}_b = \mathrm{diag}\big(p_\mathrm{ff}, p_\mathrm{ff}, p_\mathrm{ff}, -\rho_\mathrm{ff}, q_\mathrm{ff}, q_\mathrm{ff}, q_\mathrm{ff}\big) \tag{IN-10}$$

Symbols in (IN-10).

- $\mathcal{T}_\mathrm{ff}{}^a{}_b$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- p_ff: The pressure of the fold fluid along ordinary space. Unit: GeV^4. Value: $-\lambda_\mathrm{F} = -32HV_*/(5\kappa)$. Status: ASSUMED (the fluid), DERIVED (the value).
- ρ_ff: The energy density of the fold fluid. Unit: GeV^4. Value: $\lambda_\mathrm{F} = 32HV_*/(5\kappa)$ (its tension). Status: ASSUMED (the fluid), DERIVED (the value).
- q_ff: The pressure of the fold fluid along the extra times: minus its tension plus the pressure q_hold that holds the volume of the extra times fixed. Unit: GeV^4. Value: $-\lambda_\mathrm{F} + q_\mathrm{hold}$; $-32HV_*/(5\kappa)$ at the static block. Status: ASSUMED (the fluid), DERIVED (the value).

The fold: the reduction of the terms of the fold to four-dimensional fields with the volume V_* of the extra times and the factor $\mathcal{Z}$ (IN-8); the Einstein-Hilbert term that the fields of the fold may induce (IN-9), which with the corrected weights is optional, because the Gauss-Bonnet and cubic terms of the bulk give the kink of the metric an Einstein-Hilbert term of its own: the fold obeys the Friedmann equations of general relativity exactly when its extra times are frozen, with $8\pi G_\mathrm{N} = \kappa_7/V_*$ and $\kappa_7 = 15\kappa H/16$, and its gravitational waves travel at the speed of light at every wavelength; and the fold fluid (IN-10, with the values FE-65), the thin source that the kink of the warp factor requires: a pure tension λ_F at the static block, like the brane of Randall and Sundrum, whose excess over λ_F (together with the vacuum energy of the fields) is the dark energy, and the pressure q_hold along the extra times that holds their volume fixed (without it Newton's constant and every coupling of the fold would change far beyond the observed bounds). Status: ASSUMED (the fold fluid and its holding pressure), DERIVED (the reduction, the gravity of the fold, the values of the fold fluid), ESTIMATE (the induced term).

$$\begin{aligned} &p_\mathrm{ff} = -\rho_\mathrm{ff}, \qquad q_\mathrm{ff} = -\rho_\mathrm{ff} + q_\mathrm{hold}, \qquad \lambda_\mathrm{F} = \frac{32HV_*}{5\kappa}, \\ &q_\mathrm{hold} = -\Delta_\mathrm{r} - \frac{4}{5}\big(\mathsf{A}^4 + \mathsf{A}^2\mathsf{A}_1\big)\frac{HV_*}{\kappa}, \qquad \Delta_\mathrm{r} = q_\mathrm{SM} + q_\chi + q_\mathrm{P} - \big(\rho_\mathrm{ff} - \lambda_\mathrm{F}\big) + \frac{\rho - 3\mathcal{P}}{2} \end{aligned} \tag{FE-65}$$

Symbols in (FE-65).

- p_ff: The pressure of the fold fluid along ordinary space. Unit: GeV^4. Value: $-\lambda_\mathrm{F} = -32HV_*/(5\kappa)$. Status: ASSUMED (the fluid), DERIVED (the value).
- ρ_ff: The energy density of the fold fluid. Unit: GeV^4. Value: $\lambda_\mathrm{F} = 32HV_*/(5\kappa)$ (its tension). Status: ASSUMED (the fluid), DERIVED (the value).
- q_ff: The pressure of the fold fluid along the extra times: minus its tension plus the pressure q_hold that holds the volume of the extra times fixed. Unit: GeV^4. Value: $-\lambda_\mathrm{F} + q_\mathrm{hold}$; $-32HV_*/(5\kappa)$ at the static block. Status: ASSUMED (the fluid), DERIVED (the value).
- q_hold: The pressure along the extra times that the fold fluid adds to its tension to hold the volume of the extra times fixed (four-dimensional normalisation). Unit: GeV^4. Value: $-\Delta_\mathrm{r} = -0.793\,\rho_\mathrm{c}^{(0)}$ today. Status: DERIVED (the value), ASSUMED (that the fold fluid supplies it).
- λ_F: The tension of the fold: the part of the fold fluid whose pressures along ordinary space and along the extra times are minus its energy density (a vacuum energy of the seven-dimensional fold), per unit volume of ordinary space of the fold (four-dimensional normalisation). Unit: GeV^4. Value: $32HV_*/(5\kappa)$. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings $\alpha_{(p)} = w_{(p)}H^{2-2p}$ and the cosmological constant $\Lambda = -7H^2$ (the author's original value $-10H^2$ was corrected on 2026-10-25); in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- Δ_r: The mismatch of the radion condition: the sum over the matter of the fold of its pressure along the extra times and $(\rho - 3\mathcal{P})/2$ (zero for radiation). Unit: GeV^4. Value: $0.793\,\rho_\mathrm{c}^{(0)}$ today. Status: DERIVED.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); 1 at the author's vacuum, which is not a solution with the corrected weights.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless.
- q_SM: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian, because the fields of the fold do not depend on the extra times). Unit: GeV^4. Status: DERIVED.
- q_χ: The pressure of the bound states of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- q_P: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.

The values of the fold fluid with frozen extra times ($\mathsf{A} = a_4'/H$ and $\mathsf{A}_1 = a_4''/H^2$ the expansion rate and its change in units of H): its energy density and pressure along ordinary space are those of a tension, and the pressure along the extra times adds q_{hold}, which makes up the mismatch Δ_{r} of the matter of the fold (today $q_{\mathrm{hold}} = -0.793$ times the critical density). Status: DERIVED (the values), ASSUMED (that the fold fluid supplies them).

5.6 The Higgs doublet with the ad hoc vacuum

$$\mathcal{L}_{\mathrm{H}} = -\hat{h}^{\mu\nu}\big(D_\mu\Phi\big)^\dagger D_\nu\Phi - V(\Phi), \qquad V(\Phi) = \mu^2\,\Phi^\dagger\Phi + \lambda\big(\Phi^\dagger\Phi\big)^2, \qquad \lambda > 0, \quad \mu^2 < 0 \tag{IN-11}$$

Symbols in (IN-11).

- $\mathcal{L}_{\mathrm{H}}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $V(\Phi)$: The Higgs potential, $V = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$. Unit: GeV^4. Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.

$$\Phi = \frac{1}{\sqrt{2}}\begin{pmatrix}\mathcal{A}(x_4)\\ v + h\end{pmatrix}, \qquad \mathcal{A}(x_4) = v_{\mathcal{A}}\,\mathrm{e}^{-2a_{40}x_4}, \qquad a_{40} > 0, \quad x_4 \geq 0, \qquad v^2 = -\frac{\mu^2}{\lambda} \tag{IN-12}$$

Symbols in (IN-12).

- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}}\mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_{\mathrm{F}})^{-1/2}$ with the Fermi constant $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- h: The Higgs field: the fluctuation of the lower component of the Higgs doublet about v. Unit: GeV. Value: $m_h = 125.25$ GeV. Status: STANDARD.
- $v_{\mathcal{A}}$: The amplitude of almightyAdHoc at $x_4 = 0$; the author's almightyAdHoc = Exp$[-2\,a_{40}\,x_4]$ has $v_{\mathcal{A}} = 1$ in the units of his notebook. Unit: GeV. Value: 1 in the units of the author's notebook; UM-3 keeps it free (the examples take $v_{\mathcal{A}} = v$ and $10v$). Status: AUTHOR (the form), ASSUMED (a free amplitude).
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.

- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.

The Higgs doublet with the author's potential, expanded about the author's ad hoc vacuum, whose upper component almightyAdHoc decays as $\mathrm{e}^{-2a_{40}x_4}$ (the author writes Exp[$-2a_{40}x_4$]; UM-3 gives it the amplitude $v_{\mathcal{A}}$ so that it has the dimension of the Higgs field). Status: AUTHOR.

5.7 The gauge fields, the quarks and the leptons

$$
\begin{aligned}
D_\mu &= \partial_\mu + \hat{\omega}_\mu - \mathrm{i}g_3\, G_\mu^A\, t^A - \mathrm{i}g_2\, W_\mu^I\, \frac{\tau^I}{2} - \mathrm{i}g_1\, Y_f\, B_\mu, \\
\mathcal{L}_{\text{gauge}} &= -\frac{1}{4}\, G^A_{\mu\nu} G^{A\mu\nu} - \frac{1}{4}\, W^I_{\mu\nu} W^{I\mu\nu} - \frac{1}{4}\, B_{\mu\nu} B^{\mu\nu}, \\
G^A_{\mu\nu} &= \partial_\mu G^A_\nu - \partial_\nu G^A_\mu + g_3\, f^{ABC} G^B_\mu G^C_\nu, \qquad W^I_{\mu\nu} = \partial_\mu W^I_\nu - \partial_\nu W^I_\mu + g_2\, \epsilon^{IJK} W^J_\mu W^K_\nu, \qquad B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu
\end{aligned}
\tag{IN-13}
$$

Symbols in (IN-13).

- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- $\hat{\omega}_\mu$: The spin connection of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold acting on four-dimensional spinors. Unit: 1/length. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling (the strong coupling g_s). Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($\alpha_s = 0.1180$). Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- B: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- C: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- t^A: The generators of $SU(3)$ on colour triplets, $t^A = \lambda^A_{\mathrm{GM}}/2$ (the Gell-Mann matrices halved), $\mathrm{tr}(t^A t^B) = \delta^{AB}/2$. Unit: dimensionless. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling (usually g). Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- J: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- K: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- g_1: The $U(1)_Y$ gauge coupling (usually g'; not normalised for grand unification). Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- Y_f: The hypercharge of a field, the charge of its coupling $g_1 Y_f B_\mu$ to the hypercharge field in the covariant derivative (1/6, 2/3, $-1/3$, $-1/2$, -1, 0 for $Q, u_{\mathrm{R}}, d_{\mathrm{R}}, \ell, e_{\mathrm{R}}, \nu_{\mathrm{R}}$, 1/2 for Φ). Unit: dimensionless. Status: STANDARD.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.

- $\mathcal{L}_{\text{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- f^{ABC}: The structure constants of $SU(3)$, $[t^A, t^B] = \mathrm{i} f^{ABC} t^C$. Unit: dimensionless. Status: STANDARD.
- ϵ^{IJK}: The Levi-Civita symbol of three indices (the structure constants of $SU(2)$), $\epsilon^{123} = 1$. Unit: dimensionless. Status: STANDARD.

$$\mathcal{L}_{\text{f}} = \sum_{\psi} \mathrm{i}\, \bar{\psi}\, \hat{\gamma}^{\mu} D_{\mu} \psi, \qquad \psi \in \{Q, u_{\text{R}}, d_{\text{R}}, \ell, e_{\text{R}}, \nu_{\text{R}}\},$$
$$\{\hat{\gamma}^{\mu}, \hat{\gamma}^{\nu}\} = -2\hat{h}^{\mu\nu}\mathbf{1}, \qquad \bar{\psi} = \psi^{\dagger}\beta, \qquad \beta = \hat{\gamma}^{(4)}, \qquad \hat{\gamma}_5 = \mathrm{i}\,\hat{\gamma}^{(4)}\hat{\gamma}^{(1)}\hat{\gamma}^{(2)}\hat{\gamma}^{(3)},$$
$$\hat{P}_{\text{L}} = \frac{\mathbf{1} - \hat{\gamma}_5}{2}, \quad \hat{P}_{\text{R}} = \frac{\mathbf{1} + \hat{\gamma}_5}{2}, \qquad \hat{P}_{\text{L}} Q = Q, \quad \hat{P}_{\text{L}} \ell = \ell, \quad \hat{P}_{\text{R}} u_{\text{R}} = u_{\text{R}}, \quad \hat{P}_{\text{R}} d_{\text{R}} = d_{\text{R}}, \quad \hat{P}_{\text{R}} e_{\text{R}} = e_{\text{R}}, \quad \hat{P}_{\text{R}} \nu_{\text{R}} = \nu_{\text{R}} \tag{IN-14}$$

Symbols in (IN-14).

- $\mathcal{L}_{\text{f}}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_{\text{R}}, d_{\text{R}}, \ell, e_{\text{R}}, \nu_{\text{R}}$. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^{\dagger}\beta$; ψ_f the fields of the kind f. Unit: $\text{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat{\gamma}^{\mu} = \hat{\mathbf{e}}_a{}^{\mu}\hat{\gamma}^{(a)}$, $\{\hat{\gamma}^{\mu}, \hat{\gamma}^{\nu}\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu} = \hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_{μ}: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_{μ} (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^{\mu} = \hat{h}^{\mu\nu}D_{\nu}$; on an adjoint field $(D_{\mu}X)^A = \partial_{\mu}X^A + g_3 f^{ABC} G^B_{\mu} X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_{\text{L}}, d_{\text{L}})$ (colour triplets, hypercharge 1/6). Unit: $\text{GeV}^{3/2}$. Status: AUTHOR.
- u_{R}: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\text{GeV}^{3/2}$. Status: STANDARD.
- d_{R}: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\text{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_{\text{L}}, e_{\text{L}})$ (hypercharge $-1/2$). Unit: $\text{GeV}^{3/2}$. Status: AUTHOR.
- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\text{GeV}^{3/2}$. Status: AUTHOR.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\text{GeV}^{3/2}$. Status: ASSUMED.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \text{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- β: The matrix of the four-dimensional Dirac adjoint, $\beta = \hat{\gamma}^{(4)}$ (Hermitian, $\beta^2 = \mathbf{1}$). Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}_5$: The four-dimensional chirality matrix, $\hat{\gamma}_5 = \mathrm{i}\hat{\gamma}^{(4)}\hat{\gamma}^{(1)}\hat{\gamma}^{(2)}\hat{\gamma}^{(3)}$, $\hat{\gamma}_5^2 = \mathbf{1}$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_{\text{L}}$: The four-dimensional left-handed projector, $\hat{P}_{\text{L}} = (\mathbf{1} - \hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_{\text{R}}$: The four-dimensional right-handed projector, $\hat{P}_{\text{R}} = (\mathbf{1} + \hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.

$$\mathcal{L}_{\mathrm{Y}} = -\big(\bar{Q}\,\mathcal{Y}^{u}\,\tilde{\Phi}\,u_{\mathrm{R}} + \bar{Q}\,\mathcal{Y}^{d}\,\Phi\,d_{\mathrm{R}} + \bar{\ell}\,\mathcal{Y}^{e}\,\Phi\,e_{\mathrm{R}} + \bar{\ell}\,\mathcal{Y}^{\nu}\,\tilde{\Phi}\,\nu_{\mathrm{R}}\big) + \mathrm{h.c.}, \qquad \tilde{\Phi} = \mathrm{i}\tau^{2}\Phi^{*} \tag{IN-15}$$

Symbols in (IN-15).

- $\mathcal{L}_{\mathrm{Y}}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_{\mathrm{L}}, d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^{u}$, $\mathcal{Y}^{d}$, $\mathcal{Y}^{e}$, $\mathcal{Y}^{\nu}$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^{2}\Phi^{*}$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- u_{R}: The right-handed up-type quarks u, c, t (singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- d_{R}: The right-handed down-type quarks d, s, b (hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_{\mathrm{L}}, e_{\mathrm{L}})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- τ^{I}: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- *: Complex conjugation of a number, or of each entry of a matrix or column. Status: STANDARD.

The gauge fields of $SU(3)_{\mathrm{C}} \times SU(2)_{\mathrm{L}} \times U(1)_{Y}$ on the fold (IN-13), the three generations of quarks and leptons with three right-handed neutrinos (IN-14), and their Yukawa couplings to the Higgs doublet (IN-15). With the right-handed neutrinos the gauge anomalies still cancel (DERIVED, `um3_fields.py`), and the neutrino Yukawa couplings give the neutrinos Dirac masses, whose values are chosen to reproduce the measured mass differences with the lightest mass zero (ASSUMED). Status: STANDARD (UM-3 adds the right-handed neutrinos).

5.8 The portals of dirac16complex

$$\begin{aligned}\mathcal{L}_{\mathrm{P}} = & -c_{\Phi}\,\big(\Phi^{\dagger}\Phi\big)\,\bar{\Psi}P_{\mathrm{L}}\Psi - \sum_{X}\Big[\frac{1}{4}\big(c_X^{G}\,G^{A}_{\mu\nu}G^{A\mu\nu} + c_X^{W}\,W^{I}_{\mu\nu}W^{I\mu\nu} + c_X^{B}\,B_{\mu\nu}B^{\mu\nu}\big)\,\bar{\Psi}P_X\Psi \\ & + \frac{d_X}{2}\,B_{\mu\nu}\,\mathcal{D}_X^{\mu\nu} + \sum_{f} c_X^{f}\,J_f^{\mu}\,j_{X\mu}\Big], \\ \mathcal{D}_X^{\mu\nu} = \mathrm{i}\,\bar{\Psi}\,\Sigma^{\mu\nu}P_X\Psi, & \qquad j_X^{\mu} = \mathrm{i}\big(\bar{\Psi}P_X D^{\mu}\Psi - (D^{\mu}\bar{\Psi})P_X\Psi\big), \qquad J_f^{\mu} = \bar{\psi}_f\,\hat{\gamma}^{\mu}\psi_f\end{aligned} \tag{IN-16}$$

Symbols in (IN-16).

- $\mathcal{L}_{\mathrm{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.
- c_{Φ}: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^{\dagger}\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^{G}: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.

- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c^W_X: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W^I_μ, $W^I_{\mu\nu}$: The gauge field of $SU(2)_L$ on the fold, $W^1_\mu, W^2_\mu, W^3_\mu$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c^B_X: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_μ, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}^{\mu\nu}_X$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}^{\mu\nu}_X = \mathrm{i}\bar\Psi\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c^f_X: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c^ℓ_X and c^ν_X are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J^μ_f: The fermion-number current of the Standard-Model multiplets of kind f, $J^\mu_f = \bar\psi_f\hat\gamma^\mu\psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- j^μ_X: The convective current of the half X of dirac16complex, $j^\mu_X = \mathrm{i}(\bar\Psi P_X D^\mu\Psi - (D^\mu\bar\Psi)P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu\mathbf{e}_b{}^\nu\Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- ψ, $\bar\psi$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar\psi = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat\gamma^\mu = \hat{\mathbf{e}}_a{}^\mu\hat\gamma^{(a)}$, $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.

The new interactions of UM-3: every Standard-Model field couples at the fold to the left-handed and to the right-handed half of dirac16complex, each with its own coupling, through gauge-invariant operators: the Higgs portal (left-handed half only), the gauge-kinetic portals of the gluons, W and B, a dipole portal to B, and current portals between the fermion-number current of every multiplet of the Standard Model and the convective currents of the two halves. Each is invariant under the gauge group of the Standard Model and contains dirac16complex only in bilinears, so the charge of dirac16complex is conserved and its lightest states cannot decay into the Standard Model alone. The charge is indefinite (a Krein form); quantised with positive energies, every quantum carries a positive energy, so the vacuum is stable (whether a gas of the bound particles is classically stable is not decided); but the modes of positive Krein charge are ghosts (negative norm), and every coupling of the fold that distinguishes the two chiralities, the Higgs portal of the left-handed half among them, creates them together with physical particles: with these portals UM-3 is not a unitary quantum theory, and a unitary UM-3 would need couplings of the fold that do not distinguish the chiralities (necessary, not shown to be sufficient). Status: INVENTED (the portals), DERIVED (the stability of the vacuum and the ghosts, `um3_krein.py`), OPEN (the stability of the gas, a unitary coupling).

6 The parameters

parameter	meaning	value	status
H	the author's constant: the period $\pi/(6H)$ of x_0, the warp factor, the Lovelock weights	free; the bound states of dirac16complex need $H \ll q_\mathrm{b}$, where q_b is the inverse width of their profile $\mathrm{e}^{-q_\mathrm{b}\lvert\sigma\rvert}$ along the hidden direction	AUTHOR
$a_4(x_4)$, $b_4(x_4)$	the author's functions of the time	frozen extra times, $\mathrm{e}^{-a_4}b_4$ constant (b_4 proportional to e^{a_4}), with any expansion history $a_4(x_4)$: an inflation of the fold in the primordial era, the expansion of general relativity later; the author's vacuum $a_4 = Hx_4$ with constant b_4 is not a solution with the corrected weights	AUTHOR, DERIVED
$\alpha_{(p)}$, Λ	Lovelock weights and cosmological constant of the bulk	$\alpha_{(p)} = H^{2-2p}w_{(p)}$ with $w = (1, 1/20, 1/360)$, $\Lambda = -7H^2$ (the author's original values $(1, 1/32, 1/1152)$ and $-10H^2$ made gravitons faster than light)	AUTHOR (the form), DERIVED (the values)
$\kappa = 8\pi G_8$	the gravitational coupling of the bulk	free	ASSUMED
κ_7	the gravitational coupling of the fold, from the Gauss-Bonnet and cubic terms	$15\kappa H/16$; $8\pi G_\mathrm{N} = \kappa_7/V_*$ (plus $1/\kappa_\mathrm{ind}$ in $1/(8\pi G_\mathrm{N})$ if the fields of the fold induce a term)	DERIVED
κ_ind	the coupling of an induced Einstein-Hilbert term of the fold	optional; not fixed by UM-3	ESTIMATE
V_*	the physical volume of the extra times at the fold	free; Newton's constant fixes $15\kappa H/(16V_*)$	ASSUMED
ξ_H	the coupling of the Higgs doublet to the curvature of the fold	free; about 10^4 if the Higgs field drives an inflation of the fold (ESTIMATE)	ASSUMED
ρ_ff, p_ff, q_ff	the fold fluid (energy density, pressure along ordinary space, pressure along the extra times)	a tension: $p_\mathrm{ff} = -\rho_\mathrm{ff}$; with the vacuum energy ρ_vac of the fields, $\rho_\mathrm{ff} + \rho_\mathrm{vac} = \lambda_\mathrm{F} + \rho_\Lambda$, $\lambda_\mathrm{F} = 32HV_*/(5\kappa)$, $\rho_\Lambda = (2.2\ \mathrm{meV})^4$ the dark energy; $q_\mathrm{ff} = -\rho_\mathrm{ff} + q_\mathrm{hold}$, $q_\mathrm{hold} = -0.793$ times the critical density today	ASSUMED (its values DERIVED)
m_Ψ	the bulk mass of dirac16complex	free	RECORD

parameter	meaning	value	status
λ_Ψ	the self-coupling of dirac16complex	free	RECORD
c_Φ, c_X^G, c_X^W, c_X^B, d_X, c_X^f	the portal couplings ($X =$ L, R)	free; direct detection and the effective number of neutrino species N_{eff} (the energy density of the relativistic species besides the photons, in units of one neutrino species) constrain them	INVENTED
v, λ, μ^2	the Higgs vacuum and potential	$v = 246.22$ GeV, $\lambda = 0.129$, $\mu^2 = -\lambda v^2$	AUTHOR, STANDARD
$v_{\mathcal{A}}$, a_{40}	amplitude and rate of the ad hoc vacuum $\mathcal{A} = v_{\mathcal{A}} \mathrm{e}^{-2a_{40}x_4}$	the author's amplitude is 1 in the units of his notebook; UM-3 keeps $v_{\mathcal{A}}$ free (the examples take v and $10v$); $a_{40} > 0$; nucleosynthesis needs $a_{40} > \ln(v_{\mathcal{A}}/(0.1v))/(2\text{ s})$	AUTHOR, ASSUMED (a free amplitude)
g_1, g_2, g_3	the gauge couplings	the measured values	STANDARD
$\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$	the Yukawa matrices	the measured masses and mixings (Particle Data Group); neutrinos (the global fit NuFIT 5.2 of the oscillation data, November 2022, normal ordering, the fit without the Super-Kamiokande atmospheric data): the differences of the squared masses of the three mass states $\Delta m^2_{21} = 7.41 \times 10^{-5}$ eV^2 and $\Delta m^2_{31} = 2.511 \times 10^{-3}$ eV^2, the mixing angles $\theta_{12} = 33.40°$, $\theta_{13} = 8.54°$, $\theta_{23} = 49.14°$; the lightest mass 0 and the Dirac phase 0. The newer fit NuFIT 6.0 (2024) changes the results of UM-3 by at most 1.5 per cent; the Dirac phase (212 degrees in NuFIT 6.0) drops out of every result of UM-3; DESI DR2 with the CMB and NuFIT 6.0 bounds the lightest mass below 23 meV	STANDARD, ASSUMED (the lightest mass; the phase is irrelevant, DERIVED)

parameter	meaning	value	status
$A_{\rm s}$, $n_{\rm S}$, $\tau_{\rm reion}$	the primordial spectrum and the optical depth	Planck 2018: $\ln(10^{10}A_{\rm s}) = 3.044$, $n_{\rm S} = 0.9649$, $\tau_{\rm reion} = 0.0544$; they need an inflation of the fold with frozen extra times, which with the corrected weights is the inflation of general relativity (its inflaton, possibly the Higgs field, is not specified)	ASSUMED (inputs; the inflaton OPEN)